

URAKAWA, Japan

WMO: 474260

Lat: 42.1625N

Lon: 142.7769E

Elev: 126

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.5	17.8	1.2	5.9	18.0	4.2	6.8	19.8	40.3	30.6	34.8	29.6	10.3	310	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.4	76.7	72.7	74.8	71.6	73.3	70.5	73.7	75.7	72.4	74.2	71.0	72.8	8.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.0	123.0	75.2	71.7	117.6	73.9	70.3	111.9	72.5	37.5	75.9	36.3	74.3	35.0	72.9	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.1	27.0	22.8	12.0	80.4	3.2	2.8	9.7	82.4	7.8	84.0	6.0	85.6	3.7	87.6		
(5)			9.9	75.4	3.1	1.9	7.6	76.8	5.8	77.9	4.0	79.0	1.7	80.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	27.4	27.8	33.5	41.4	49.8	56.5	64.3	68.4	64.3	54.1	42.9	32.4
	DBStd	14.98	4.75	4.81	4.69	4.67	4.66	4.58	4.40	4.05	4.78	5.30	6.72	5.85
	HDD50	2958	701	621	512	262	61	5	0	0	0	23	226	547
	HDD65	6780	1166	1041	977	708	471	255	65	15	70	338	662	1012
	CDD50	1863	0	0	0	4	55	201	443	569	428	149	14	0
	CDD65	212	0	0	0	0	0	1	44	119	48	0	0	0
	CDH74	255	0	0	0	0	0	0	34	157	64	0	0	0
	CDH80	6	0	0	0	0	0	0	1	4	1	0	0	0
Wind	WSAvg	9.6	11.5	11.3	11.2	9.4	8.3	7.0	6.5	6.7	8.6	10.4	11.9	12.2
Precipitation	PrecAvg	44.6	1.6	1.3	2.1	3.3	4.6	4.3	5.5	6.1	5.5	4.2	3.5	2.5
	PrecMax	59.1	3.4	3.4	5.7	8.9	9.0	10.7	14.9	16.9	11.5	9.4	8.4	5.8
	PrecMin	25.4	0.2	0.1	0.3	0.4	0.7	1.0	0.7	1.1	1.7	0.9	0.4	0.6
	PrecStd	7.3	0.8	0.8	1.1	1.6	2.0	2.4	3.1	3.5	2.3	2.0	1.6	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	43.5	49.1	59.5	65.1	70.1	77.2	79.9	79.1	69.0	61.0	51.7
		MCWB	39.5	41.3	44.3	50.2	57.2	64.2	72.9	74.7	74.1	63.6	57.0	49.1
	2%	DB	39.2	39.5	45.6	54.1	61.8	67.1	74.5	77.5	75.6	66.5	58.2	48.1
		MCWB	35.9	36.5	41.8	48.9	55.5	62.7	71.4	73.3	71.8	61.9	54.6	45.2
	5%	DB	36.9	37.4	43.0	51.2	59.4	65.1	72.4	75.7	73.7	64.6	55.9	44.8
		MCWB	33.4	34.2	39.5	47.0	54.5	61.6	69.9	72.2	70.1	59.7	52.2	41.2
	10%	DB	34.9	35.6	40.8	48.7	57.2	63.3	70.6	74.1	71.9	62.7	53.6	41.6
		MCWB	31.2	32.4	37.5	44.9	53.1	60.4	68.3	71.3	68.3	58.0	49.6	37.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.3	41.5	46.1	52.7	59.8	65.8	73.8	75.5	75.3	65.9	58.6	49.6
		MCDB	41.9	42.7	48.3	56.0	62.8	68.0	76.0	78.5	77.4	67.4	60.3	51.3
	2%	WB	36.8	37.4	42.6	49.8	57.3	63.8	72.1	74.2	73.0	62.8	55.3	45.6
		MCDB	38.7	39.0	45.0	53.1	60.1	66.1	74.0	76.5	74.8	65.3	57.5	47.9
	5%	WB	34.0	34.9	40.0	47.6	55.5	62.2	70.3	73.0	71.0	60.9	52.9	41.6
		MCDB	36.3	36.8	42.5	50.6	58.2	64.4	71.9	74.9	73.0	63.7	55.5	44.5
	10%	WB	31.7	32.8	37.9	45.4	53.7	60.8	68.7	71.9	69.0	58.7	49.9	37.7
		MCDB	34.5	35.3	40.5	48.3	56.6	62.9	70.3	73.6	71.4	62.0	53.2	41.0
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.8	10.6	10.8	10.7	9.3	8.2	8.4	11.0	12.6	11.8	10.7
		MCDBR	11.6	11.2	12.5	13.9	13.7	11.8	9.6	9.5	11.0	12.8	13.2	13.7
		MCWBR	10.8	10.6	10.7	10.4	9.3	8.1	6.7	6.4	8.0	10.4	12.8	13.4
	5% WB	MCDBR	11.6	11.5	12.2	13.3	12.2	10.6	9.2	8.6	9.6	11.8	13.7	13.9
		MCWBR	11.9	11.7	11.1	10.5	9.4	7.9	6.7	6.1	7.6	10.8	14.3	14.4
Clear-Sky Solar Irradiance	taub	0.339	0.385	0.461	0.515	0.532	0.503	0.496	0.474	0.416	0.400	0.382	0.355	
	taud	2.172	2.031	1.891	1.838	1.890	2.052	2.148	2.225	2.343	2.249	2.191	2.155	
	Ebn at Noon	246	250	244	241	241	249	249	251	258	245	227	226	
	Edh at Noon	31	43	56	63	62	52	47	43	35	34	31	30	
All-Sky Solar Radiation	RadAvg	563	838	1198	1499	1599	1577	1428	1323	1175	911	573	451	
	RadStd	33	68	86	86	141	145	116	138	109	97	53	46	38

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page